SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability.* (BL3, BL4)

Activity Tool : Mock Interview (Low)
Assessment Tool Weightage: 30%

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QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'	BL3 – Apply	**5**
2	Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'	BL4 – Analyze	**5**
3	Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'	BL3 – Apply	**5**
4	Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'	BL3 – Apply	**5**
5	Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'	BL4 – Analyze	**5**
6	Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'	BL4 – Analyze	**5**
7	Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'	BL4 – Analyze	**5**

8	Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'	BL3 – Apply	**5**
9	Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'	BL4 – Analyze	**5**
10	Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'	BL4 – Analyze	**5**

Code of Conduct:

I **VENNA MEENAKSHI REDDY(192471048)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Venna Meenakshi Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated

Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

1. Interviewer: 'Explain the steps of query processing in DBMS. How does a query optimizer decide the best execution plan?'

Query processing has 4 main steps:

1. Parsing – Checks the SQL query.
2. Translation – Converts it into an internal form.
3. Optimization – Finds the best way to execute the query.
4. Execution – Executes the selected plan and gives the result.

The optimizer chooses the plan with the lowest estimated cost based on things like indexes, number of records, joins, and disk I/O.

2. Interviewer: 'What is the difference between heuristic optimization and cost-based optimization? When would you use each?'

Heuristic optimization uses predefined rules.

Example: Perform selection before join to reduce data.

Cost-based optimization calculates the cost of different plans and chooses the cheapest one.

- Heuristic = rules
- Cost-based = lowest cost

3. Interviewer: 'How would you implement database connectivity in a real project? Explain the difference between JDBC and ODBC.'

JDBC is used to connect Java applications to databases.

ODBC is a general database connectivity interface that can be used by applications in different programming environments.

JDBC example: Java application → JDBC → MySQL database.

4. Interviewer: 'What are ACID properties? Give a real-world example where each property is critical.'

ACID has four properties:

- **Atomicity** – Transaction happens completely or not at all.
- **Consistency** – Database remains correct.
- **Isolation** – Transactions do not interfere incorrectly with each other.
- **Durability** – Once committed, data is permanently saved.

Example: In a bank transfer, money must be deducted from one account and added to another correctly.

5. Interviewer: 'Explain concurrency control problems: dirty read, lost update, and unrepeatable read. How does 2PL solve them?'

Dirty Read: Reading uncommitted data.

Lost Update: One transaction's update is overwritten by another.

Unrepeatable Read: Reading the same data twice gives different values.

2PL (Two-Phase Locking) uses locks to control transactions.

It has:

- **Growing phase** – Acquire locks.
- **Shrinking phase** – Release locks.

It helps maintain conflict serializability.

6. Interviewer: 'What is deadlock in DBMS? Describe the deadlock detection algorithm and at least two prevention strategies.'

A deadlock occurs when two transactions wait for each other forever.

Example:

T1 holds A → waits for B

T2 holds B → waits for A

Detection:

The DBMS creates a Wait-For Graph.
If there is a cycle, a deadlock exists.

Prevention:

- **Wait-Die**
- **Wound-Wait**

7. Interviewer: 'Compare Immediate Update and Deferred Update recovery. In what scenarios is each preferred?'

Immediate Update:

- Database can be updated before commit.
- Uses UNDO/REDO during recovery.

Deferred Update:

- Database is updated only after commit.
- Mainly uses REDO.

Simple difference:

Immediate = update before commit
Deferred = update after commit

8. Interviewer: 'Explain Shadow Paging. What are its advantages and limitations compared to log-based recovery?'

Shadow Paging is a recovery technique that keeps two page tables:

- **Shadow page table** – Original data.
- **Current page table** – Modified data.

If the transaction fails, the DBMS uses the shadow page table to restore the old data.

Advantage: Simple and fast recovery.

Limitation: Requires extra page management and can cause fragmentation.

9. Interviewer: 'What is a serializability schedule? How do you test for conflict serializability using a precedence graph?'

A schedule is serializable if its result is the same as executing transactions one after another.

To check conflict serializability:

1. Create a node for each transaction.
2. Draw an arrow for conflicting operations.
3. Check for a cycle.

No cycle → Conflict serializable

Cycle → Not conflict serializable

10. Interviewer: 'Describe the Two-Phase Locking protocol variations: Basic 2PL, Strict 2PL, and Rigorous 2PL. Which is most commonly used and why?'

Basic 2PL:

- Growing phase → acquire locks.
- Shrinking phase → release locks.

Strict 2PL:

- Holds write locks until commit/abort.

Rigorous 2PL:

- Holds both read and write locks until commit/abort.

Most commonly used:

Strict 2PL is commonly used because it provides good concurrency and makes recovery easier.